Algorithms, 2026 Fall, Homework 5
(Due: Oct 30)

October 20, 2026

required reading: section 6.5 6.9 6.10

Problem 1 : As I mentioned in the class, in a general directed graph (with possibly negative weights), the shortest path from s to t may not be well-defined, if there is a negative cycle in the graph that is reachable from s, and t is reachable from the negative cycle (i.e., there is a path from s to a negative cycle, and a path from the negative cycle to t). In this case, we can travel around the negative cycle to make the shortest path distance arbitrarily negative. Try to design an efficient algorithm to detect this situation. That is, given a directed graph $G(V, E)$ with possibly negative weights, and source node s, for any every other node t, the algorithm should output a shortest path from s to t if it exists, or report that the shortest distance from s and t is not well-defined. You algoirthm should run in $O(mn)$ time, where m is the number of edges and n is the number of nodes. hint recall that a directed graph can be represented as a DAG of strongly connected components.

Problem 2 : There is a pile of n sticks. S is a set of positive integers. Two players A and B take turns removing s sticks for some $s \in S$. A starts first. The players who removes the last stick wins. Design a pseudo-polynomial time algorithm (i.e., polynomial in n) that decides which player has a winning strategy (i.e., no matter how the opponent plays, the player can take certain moves to win the game.)

Problem 3 : We define an interval graph G as follows: Suppose we have a collection of intervals on the real line. Each node of G represent an interval and two nodes are connected by an edge if two corresponding interval overlap. We have shown that the maximum independent set problem (an independent set is a subset of nodes that are not connected by any edge) in interval graphs (the first interval scheduling problem) (if we know the interval representation) is polynomial time solvable (in fact $O(n \log n)$ time). In this problem, we consider the coloring problem in such graphs. In the coloring problem, we assign each vertex a color such that no edge has two endpoints colored by the same color.

We would like to use as few colors as possible. Consider the following algorithm which runs in iterations. In each iteration, we find a maximum independent set, assign a new color to all vertices in this set, and delete all vertices in the set. Prove this algorithm finds the optimal coloring, or give a counter example to show the algorithm is not optimal.

Problem 4 : (Traveling salesman path for the set of vertices of a convex polygon) We are given a convex polygon with n vertices $\{1, 2, \ldots, n\}$ in the 2-d plane. The coordinates of these vertices are given as input in clockwise order. Design an polynomial time algorithm to find a shortest traveling salesman path starting at vertex 1. (A traveling salesman path is a path that traverses each vertex exactly once.)

Problem 5 : (1) Show that the treewidth of a tree is 1.

(2) Show that the treewidth of (a tree plus one edge) is at most 2. (you need to build a tree decomposition for a graph)

Problem 6 : Suppose $G(V, E)$ is the union (of edges) of two trees defined on V. How large can the treewidth of G be? (try to make it as large as possible).

Problem 7 : The definition of a chordal graph is the following (why the name is so called): A chordal graph is an undirected graph with the following property: every cycle of length four and greater has a chord (a chord is an edge that is not part of the cycle but joins two vertices of the cycle).

Prove the following statement. We are given a tree decomposition T of the graph G. If we make every bag in T a clique (a complete subgraph) by adding edges to G, the resulting graph is a chordal graph.

Problem 8 : In this problem, we will prove the separator lemma I mentioned in the class. Suppose $G(V, E)$ is an undirected graph and $T = TD(G)$ is a tree decomposition of G. Suppose (b, b') is an edge in T, where b and b' are two bags. Show $b \cap b'$ is a separator of G in the following sense: Suppose we remove edge (b, b') from T. It is easy to see that T becomes disconnected. Let T_1, T_2 be the two resulting components. Now consider two different vertices $v, u \in V$ $(u, v \notin b \cap b')$. Suppose $v \in b_1 \in T_1$ and $u \in b_2 \in T_2$ where b_1 is a bag in T_1 and b_2 is a bag in T_2.

Prove that if we remove all vertices in $b \cap b'$ from G, u is disconnected from v. (this is equivalent to the statement I wrote in the blackboard.)

Problem 9 : In the minimum coloring problem, we need to assign a color to each node of the given undirected graph, such that any two adjacent nodes are assigned different colors. The goal is to use as few colors as possible. Solve the minimum coloring problem on graphs with bounded treewidth (i.e., the treewidth of the graph is $O(1)$) in polynomial time. You can assume a tree decomposition with constant treewidth is given. (write down the dynamic programming solution. Do not use Courcelle's theorem as a black box.)

Problem 10 : Another definition of the chordal graph is the following.

A chordal graph can be represented as the intersection graph of subtrees on a tree: each subtree corresponds to a vertex; two subtree intersects (they touch

on any point) iff the two corresponding vertices are joined by an edge.

In this problem, we prove this definition is equivalent to the previous definition. We prove the easier direction: if a graph can be represented as the intersection graph of subtrees on a tree, it is chordal.